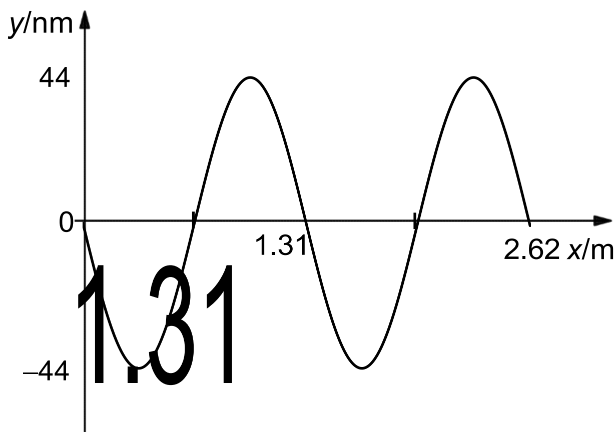
Qn	Suggested Answer
5	
(a)(i)	Definition of speed: speed $= \dfrac{\text{distance}}{\text{time}}$, $v = \dfrac{d}{t}$ Definition of frequency: frequency $= \dfrac{1}{\text{period}}$, $f = \dfrac{1}{T}$ In one cycle of a wave, the time taken is one period and the distance covered is one wavelength.
	Hence, $v = \dfrac{\lambda}{T}$ and $v = f\lambda$ (shown)
(a)(ii)	$\lambda = \dfrac{v}{f}$ $= \dfrac{344}{262}$
	$\lambda = 1.31\text{ m}$ (3sf)
(a)(iii)	 Negative sine graph for 2 cycles.
(b)(i)	The principle of superposition states that the net displacement at a given place and time caused by two or more waves which transverse the same space and meet
	is the vector sum of the displacement which would have been produced by the individual waves separately at that position and instant of time.
(b)(ii)	Let the distance between the finger and the far end be L. $L = \dfrac{\lambda}{2} = \dfrac{v}{2f}$
	change in d = change in L change in $L = \dfrac{v}{2}\left(\dfrac{1}{f_1} - \dfrac{1}{f_2}\right)$
	change in $L = \dfrac{425}{2}\left(\dfrac{1}{247} - \dfrac{1}{262}\right)$
	$= 0.0493\text{ m}$ (3sf)
(c)(i)	Diffraction is the bending/spreading of a wave into its geometrical shadow when it is incident on an edge of an obstacle or aperture/opening.

Qn	Suggested Answer
(c)(ii)	The first order bright fringes will • be <u>brighter</u> / have <u>greater intensity</u>. • be spaced <u>further</u> from the central maximum. • the <u>width</u> of the fringes will <u>decrease</u>.
(c)(iii)	When only slits A and C are covered, the fringe pattern is that of a single slit diffraction pattern through slit B. When only slit B is covered, the fringe pattern is that of a double slit pattern through slits A and C.
	<u>Single Slit (ss) Diffraction</u> $\frac{z_{ss}}{D}=\frac{\lambda}{b}$ so $b=\frac{\lambda D}{z_{ss}}$ <u>Double Slit (ds) Interference</u> $\frac{z_{ds}}{D}=\frac{\lambda}{a}$ so $a=\frac{\lambda D}{z_{ds}}$ Where D is the distance from the slit to the screen, b is the width of each slit, a is the separation between slits A and C and λ is the wavelength of incident light.
	$\frac{a}{b}=\frac{z_{ss}}{z_{ds}}$
	Hence, for the single slit diffraction pattern, the distance from P to the first minimum is $z_{ss}=0.01\text{ mm}$. For the double slit interference pattern, the distance from P to the first maximum is $z_{ds}=\frac{0.01}{3.5}\text{ mm}$
	$\frac{a}{b}=\left(\frac{0.01}{0.01/3.5}\right)$
	= 3.5
(c)(iv)	Width of central bright maximum of the diffraction pattern through slit C is the same as the width of central bright maximum of the diffraction pattern through slit B.
	0.02 mm

Qn	Suggested Answer
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